

Nfs Mental Health Platform

Non-Functional Requirements - Performance, Security, Usability, and Reliability Criteria

This document defines the non-functional requirements for the Nfs Mental Health Platform. These requirements describe how the system should perform, how secure it should be, how easy it should be to use, and how reliable it should be during normal operation and testing.

Performance Requirements

ID	Requirement	Acceptance / Validation Method
NFR-P01	Average API response time shall be less than 500ms in the testing/local environment.	API performance test
NFR-P02	Main frontend pages shall load within 2 to 3 seconds.	Frontend page load test
NFR-P03	The system shall support 20 to 50 concurrent users in the tested scenario.	Load test
NFR-P04	The successful request rate shall be greater than 95%.	API monitoring / test report
NFR-P05	AI chatbot responses shall be optimized using caching and response-size control where possible.	AI response testing
NFR-P06	The database shall use indexes for frequently queried fields such as email, role, session date, therapist ID, client ID, and notification status.	Database review

Security Requirements

ID	Requirement	Acceptance / Validation Method
NFR-S01	Passwords shall be stored using secure hashing and never as plain text.	Code and database inspection
NFR-S02	The system shall enforce authentication and role-based authorization.	Access control testing
NFR-S03	Clients shall not access other clients private data.	Authorization test
NFR-S04	Therapists shall only access assigned sessions, conversations, and permitted notes.	Authorization test
NFR-S05	Admins shall have management privileges, and all sensitive admin actions shall be logged.	Audit log verification
NFR-S06	Sensitive data such as therapy notes, chats, and AI logs shall be protected using encryption or secure storage methods.	Security review
NFR-S07	AI logs shall be anonymized when used for improvement or analysis.	Data privacy review

NFR-S08	The system shall validate all inputs to reduce SQL injection, XSS, and invalid data submission risks.	Security testing
NFR-S09	The system shall support anonymous participation in group sessions using nickname or avatar.	Feature test

Usability Requirements

ID	Requirement	Acceptance / Validation Method
NFR-U01	The interface shall be simple and easy to navigate for clients, therapists, and admins.	Usability review
NFR-U02	Users shall be able to complete key tasks such as booking a session in three steps or less.	Task completion test
NFR-U03	The system shall display clear error messages for failed actions such as invalid login or unavailable time slot.	UI test
NFR-U04	The interface shall be responsive on desktop, tablet, and mobile screen sizes.	Responsive design test
NFR-U05	The platform shall provide privacy-friendly options such as anonymous group participation.	Feature test
NFR-U06	The user experience shall be culturally suitable for users in the Middle East.	Design review

Reliability Requirements

ID	Requirement	Acceptance / Validation Method
NFR-R01	The system shall prevent duplicate bookings for the same therapist and time slot.	Booking conflict test
NFR-R02	The system shall save confirmed bookings, questionnaire responses, recommendations, chat messages, and wellbeing records correctly.	Data persistence test
NFR-R03	The system shall show unavailable status when a group session is full.	Group session test
NFR-R04	The system shall handle errors gracefully without crashing the whole platform.	Exception handling test
NFR-R05	The system shall keep logs for debugging, monitoring, and auditing.	Log review
NFR-R06	The system shall support backup and recovery plans to reduce data loss.	Backup plan review
NFR-R07	The system shall use modular architecture and early integration testing to reduce integration issues.	Architecture review

Acceptance Criteria

Performance Criteria

- Average API response time is less than 500ms during testing.
- Page load time is less than 2-3 seconds.
- Successful request rate is greater than 95%.
- The system supports 20-50 concurrent users in a tested scenario.
- Chatbot responses are optimized using caching where possible.

Security Criteria

- Users cannot access dashboards without logging in.
- Clients cannot access other clients private data.
- Therapists can only access their assigned sessions and conversations.
- Admins can manage platform data, but all sensitive actions are logged.
- Passwords are not stored as plain text.
- AI logs are anonymized when used for improvement.
- Group session users can choose nickname or avatar.

Usability Criteria

- Users can register and log in easily.
- Users can book a therapy session in three steps or less.
- Users can clearly see upcoming sessions and notifications.
- Users can join group sessions with nickname or avatar.
- Error messages are clear and understandable.
- The interface works properly on desktop and mobile screens.

Reliability Criteria

- The system prevents double booking.
- The system saves all confirmed bookings correctly.
- The system shows unavailable status when a group session is full.
- The system keeps audit logs for important actions.
- The system continues working when one non-critical module fails.
- The system supports backup and recovery.

Non-Functional Requirements Summary

- Performance: The system should respond quickly, load pages within expected time, and support the tested number of concurrent users.
- Security: The system should protect sensitive data using authentication, authorization, secure storage, anonymization, validation, and audit logs.
- Usability: The system should be simple, responsive, culturally suitable, and allow users to complete key tasks easily.
- Reliability: The system should prevent duplicate bookings, handle errors gracefully, save data correctly, keep logs, and support backup and recovery.

Source basis: Nfs project proposal, user stories and use cases, stakeholder analysis, KPIs, risk assessment, project plan, and task assignment documents.